Integrated Quantum Cryptography for Secure Data and Authentication

MINOR PROJECT REPORT

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

Department of Computer Science and Engineering

Submitted to

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)

Submitted by

Naman Aggarwal (24U020023 V Semester)

Under the supervision of

Dr. Saurabh Jain

Assistant Professor

(Computer Science and Engineering)

November 2026

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
BHOPAL (M.P.)

CERTIFICATE

This is to certify that the work embodied in this report entitled **"Integrated Quantum Cryptography for Secure Data and Authentication"** has been satisfactorily completed by **Naman Aggarwal(24U020023)**. It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering**, **Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Technology during the academic year 2026-27.

Date : 2 November, 2026

Dr. Saurabh Jain
Assistant Professor,
Department of CSE,
IIIT Bhopal (M.P.)

Dr. Rahul Chaurasia,
Department of CSE,
IIIT Bhopal (M.P.)

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL (M.P.)

DECLARATION

We hereby declare that the following minor project report entitled "Integrated Quantum Cryptography for Secure Data and Authentication" presented here is in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Saurabh Jain**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

Naman Aggarwal (24U020023)

ACKNOWLEDGEMENT

We, Naman Aggarwal, would like to thank to everyone who helped and directed us in successful completion of our minor project "Integrated Quantum Cryptography for Secure Data and Authentication"

First of all we would like to thank our esteemed Director, Prof. Dr. Ashutosh Kumar Singh for giving us the opportunity to undertake our project by giving us the resource and facilities for doing the project. His vision and enthusiasm has kept us motivated through out the completion of the project.

We are highly thankful to our Project Guide, Dr. Saurabh Jain for her round the clock guidance, suggestions, and valuable advice through out the project. We, grateful to her knowledge and inputs, which enabled us to make our project more effective and we are motivated by her support.

We would like to thank our Department Coordinator Dr. Rahul Chaurasia, for the support and encour- agement given while working on this project.

Besides, we would like to thank all the faculties of the CSE for their precious guidance and support in clarifying our doubts and providing the right direction.

Thanks to our friends for their constant support and suggestions, and helping us through many stages of this project. Their suggestions and support kept us engaged and motivated throughout the project work.

And finally, and not the least we would like to thank our parents and families for their love, patience and encouragement and for the moral support throughout the project. Their understanding towards us and support were the prime motivator to what we are today.

We thank you all dearly for all the help that you gave us directly and indirectly to make this project a success. Without your help and support this project would not have taken place.

Naman Aggarwal [24U020023]

Area of Work

This project aims to explore how quantum information science can be applied to develop secure communication. Due to the increased number of devices being connected within modern power systems, ensuring secure communication between entities is becoming a key research area. For example, smart grid communication requires service providers, smart meters and other entities to communicate using operational data. This requires secure communication to ensure entities can be authenticated, any unwanted interference can be detected and data can be kept secure and reliable.

The research aims to understand how semi-quantum communication can be used for authentication and secure data transfer. The theory is based around research into quantum assisted communication for smart grid systems. Key areas of interest include quantum entanglement, Bell states, decoy state verification, mutual authentication and message authentication. The research also aims to understand how quantum state preparation and measurement can be used for the chosen authentication and communication methods.

The project has been implemented using Python with Qiskit and Qiskit Aer. This allowed a simulation to be created where the chosen protocol methods can be implemented and analysed. This included creating quantum circuits, generating and measuring Bell states, authenticating identities and securely sending data to and from each entity. Message authentication using XOR was also explored to see if the data sent was consistent. Overall this project links the theory of semi-quantum cryptography to the practical side by using quantum circuit simulation.

Table of Content

S.no	Title	Page No.
	Certificate	
	Declaration	
	Abstract	
1	Introduction	2
2	Literature Review or Survey	5
3	Problem Definition And Objective	9
4	Proposed Methodology And Work	10
5	Proposed Algorithm	15
6	Proposed flowchart/ DFD/ Block Diagram	20
7	Implementation (Tools & Technology Used)	25
8	Result Analysis	29
9	Conclusion & Future Scope	36
10	References	37

List of Figures

Fig. No.	Description	Page No.
6.1	Proposed System Block Diagram	20
6.2	Proposed System Flowchart	21
6.3	DFD Level 0 – Context Diagram	22
6.4	DFD Level 1 – Detailed Data Flow	23
8.1	Quantum Circuit for Bell-State-Based Identity Authentication	29
8.2	SP-to-SM Quantum Message Transmission Circuit	30
8.3	SM-to-SP Quantum Message Transmission Circuit	31
8.4	Noisy Single-Qubit Measurement Benchmark	32
8.5	Noisy Bell-State Measurement Benchmark	33
8.6	Noisy Teleportation Decoding Benchmark	33

List of Tables

Table No.	Description	Page No.
2.1	Overview of Key Research Contributions in Quantum Cryptography and Smart Grid Communication	7
4.1	Truth Table Representing the XOR Operation	13
5.1	Phases of the Bidirectional Data Transmission and Authentication Algorithm	15
5.2	Noisy Simulation Benchmarks and Their Evaluation Objectives	17
5.3	Configurations of the BB84 Intercept-and-Resend Experiment	17
8.1	Experimental Parameters	28
8.2	Decoy Verification and Identity Authentication Results	28
8.3	SP-to-SM Transmission Results	29
8.4	SM-to-SP Transmission Results	31
8.5	Message Authentication Results	32

Abstract

With increased reliance on communication networks, the information security aspect of smart grids becomes very important. Communication occurs between service providers and smart meters so it's very important that this information is kept secure. Traditional encryption methods can be problematic due to future quantum computers. Quantum cryptography can help with this as it uses quantum mechanics to create a secure system.

This project looks at how to use semi-quantum computations to secure the two-way information exchange and authentication processes. This project aims to look at a theory described by Lou et al [1] and investigate how to securely communicate between smart grid components. The theory investigates how to use quantum entanglement, Bell states, decoy states and measurements to create secure communications between two parties, one with less quantum capabilities than the other. This includes two-way authentication of identities, detecting eavesdropping, two-way information exchange and message authentication.

A simulation was created using Python, Qiskit and Qiskit Aer. The simulation creates quantum circuits to look at how the Bell states are created, and how they are measured. This includes looking at how authentication and information exchange works between the two parties. Message authentication was also looked at using XOR. A BB84-style intercept resend simulation was also completed.

The simulation demonstrates some parts of the protocol. It shows that the information can be recovered and messages can be authenticated. The project gives practical insight into how quantum information can be used to create secure communications. However, there is a difference between this simulation and a full implementation of the original research.

Introduction

1.1 Background

With the fast advancement of information and communication technologies, there is an increasing need for data handling. Smart grids, healthcare systems, industry systems, etc., rely on communication between interconnected devices. However, this can lead to security vulnerabilities. When transmitting data over a network there could be a risk of unauthorised access, eavesdropping, tampering and spoofing. It is therefore important to ensure that information is kept confidential, intact and genuine.

Smart grids are advanced electrical networks that incorporate traditional systems along with information and communication technologies. This means that there is communication between the electricity providers, substation and smart meters. However, as there are many interconnected devices there could be a risk of an attacker trying to gain access to sensitive information, spoofing and tampering with messages being transmitted. Secure methods are needed to ensure smart grids can communicate securely [8].

Cryptographic methods can help to keep communications secure through mathematical algorithms and computational assumptions. However, the development of quantum computing has raised concerns about the future security of some public key cryptographic systems. Shor's algorithm showed that if quantum computers are powerful enough they can solve integer factorisation and discrete logarithm problems quickly [9]. This has led to research in areas such as quantum cryptography and post-quantum cryptography.

1.2 Introduction to Quantum Cryptography

Quantum cryptography uses the principals of quantum mechanics. A significant application of this is Quantum Key Distribution (QKD) where a secret key can be created between two legitimate parties using quantum states. This is secure due to the nature of the quantum information.

A significant development in this area was the introduction of the BB84 protocol [2] by Bennett and Brassard in 1984. This uses quantum states prepared in different bases. The sender sends the quantum states and the receiver measures them using random bases. The parties then compare their bases and keep the matching results. If an eavesdropper measures the quantum states they will cause a disturbance to the quantum states which will result in errors within the system.

The security of quantum communication is related to the uncertainty principle and no cloning theorem. Wootters and Zurek showed that it is impossible to make a perfect copy of an arbitrary unknown quantum state [4]. This makes quantum information fundamentally different to classical information where it is generally possible to copy.

A significant development was the introduction of the E91 protocol by Ekert [3] in 1991. This uses quantum states that are entangled between the parties. The correlation between the measurement outcomes can be analysed using Bell's theorem to detect any possible interference. This made entanglement an important resource within quantum communication.

Theoretically, the security of QKD has been studied. The security of the BB84 protocol was shown [5] by Shor and Preskill. However, research has been carried out into the security issues associated with practical implementations [7] [10]. This shows the importance of taking into account losses during transmission, noise, device limitations and possible attacks.

1.3 Semi-Quantum Cryptography

Traditional quantum protocols may assume that each party has access to advanced quantum processes. This may not always be an option. Semi-quantum cryptography allows one party to have limited capabilities.
One party in a semi-quantum key distribution protocol only has limited operations available to them, for example they can measure/prep states in a particular basis, reflect their states or perform some other operation [6]. The other party has higher quantum operations available and explores if secure communication can be achieved when both parties do not have the same capabilities.
Semi-quantum protocols would be useful in situations where there are different technology levels within your communication. The security of the protocol will depend on its design, what allowed operations are used, authentication and attack analysis. Just because you have limited quantum capabilities does not mean the communication will be secure.

1.4 Quantum Cryptography in Smart Grid Communication

There are bidirectional communications in the smart grids. For example, between service provider and smart meter. These bidirectional communications need to be secure as the information being communicated could be altered or impersonated.
An effective communication system needs to be able to do more than just keep the information confidential. The communication system needs to authenticate the communication partners to ensure the identities of the devices are correct. The messages also need to be authenticated to ensure the information being exchanged is not altered. This is done using identity authentication to authenticate the device and message authentication to ensure the information is authentic. It is also important to be able to detect eavesdropping as measurements of the quantum states may be unauthorised.
A research paper written by Lou et al [1] entitled ‘Privacy-Preserving Bidirectional Data Transmission of Smart Grid Via Semi-Quantum Computation: On Mutual Identity and Message Authentication’ outlines how bidirectional communications can be achieved using semi-quantum methods. The protocol they propose has mutually authenticated identity and message authentication. They outline how the quantum communication protocols can be used considering that quantum states will be entangled, measurements will be taken and eavesdropping detected for devices with differing levels of quantum capability.
This research has provided the main theory behind this project. As it considers identity authentication and message authentication for bidirectional data.

1.5 Motivation and Scope of the Project

The purpose of this project is to explore how quantum information can be utilised to provide secure data transmission and authentication within communication systems. This will involve exploring specific topics within semi-quantum communications, such as: preparing Bell states, quantum measurements, verifying decoy states, mutual authentication and exchanging data.
The practical portion of this project will involve creating a Python-based simulation using Qiskit and Qiskit Aer. Circuits can be created to explore Bell state preparation, quantum measurements, specific authentication strategies, and exchanging binary data between two entities. Message authentication is also explored using XOR. Finally, this project will also include creating a BB84-inspired intercept resend simulation.
This project will only implement the specific components included in the simulation. This will give an insight into the concepts in the referenced research paper [1], however it does not fully recreate all aspects of the protocol and there is no security proof provided.
Overall, this project will develop an understanding of quantum states, entanglement, quantum measurements and authentications within secure communication. It will also give hands-on experience with creating quantum circuits with Qiskit. This will provide a basis for further research into quantum cryptography.

Literature Review

2.1 Early Research on Quantum Cryptographic Protocols

Initially research was focused upon creating secure methods of distributing secret keys in quantum cryptography. The BB84 protocol [2], developed by Bennett and Brassard in 1984, demonstrates how secret keys can be created using quantum states prepared in two bases. This allowed secret keys to be created using different quantum states and enabled detection of eavesdropping. There were many advancements made within quantum cryptography due to the BB84 protocol.

An alternative protocol was suggested by Ekert [3] in 1991. The E91 protocol uses entangled pairs of particles and the correlation between the measurement results to gain insights into how secure the quantum communication is. The E91 protocol makes use of quantum entanglement. This provides a different model for the communication compared to the BB84 protocol.

Following the theoretical protocols there was research into the security of the protocols. A simpler proof was presented for the security of BB84 by Shor and Preskill [5]. This gave a more robust mathematical background to the protocol. Later Scarani et al. [7] looked at the security of practical quantum key distribution systems. This showed the difference between the theoretical ideal system and the practical system. Due to imperfections of the devices, transmission losses and vulnerabilities from the implementation, the security of practical quantum cryptography systems must be considered.

This provided a starting point for quantum communication and encouraged researchers to look into protocols that could operate in various different circumstances.

2.2 Research Developments in Semi-Quantum Cryptography

A significant advancement in quantum cryptography is the introduction of semi-quantum key distribution, first presented by Boyer et al. [6] in 2007. This involves protocols where one party has full quantum capabilities whilst the other party only has access to limited quantum operations. This reduced the quantum processing capabilities required by one party, creating a new model of protocol design.

This has brought many opportunities for further research in quantum cryptography. A large focus of research has been on protocols involving parties with unequal quantum capabilities, not requiring every party to have complex quantum devices. This becomes relevant in environments with limited resources.

Iqbal et al. [11] provided a survey of semi-quantum cryptography in 2019. This explored how semi-quantum protocols were developed, and covered the theory, security and possible applications of semi-quantum protocols. It also covered the various methods involved in limited quantum parties, showing how semi-quantum cryptography has become a wider area of research.

The research above provides important background knowledge on the communication model used within this project. It explains why protocols using limited quantum operations are a

worthy area of research for differing technological capabilities between communication parties.

2.3 Research on Authentication and Smart Grid Communication

Authentication is another key requirement when considering smart grid communications. Qasaimeh, Turab and Al-Qassas have conducted a review of authentication methods within smart grid environments [12]. The work discusses different methods of authentication that have been considered. It also outlines how important it is to authenticate entities within a communication while taking into account the limited resources of devices used.

The work by Kong [13] considers quantum key distribution protocols with regard to securing smart grid communications. The study highlights how applicable QKD is to smart grids and the challenges that are faced such as the requirements of the quantum communication system, transmission issues and the large amount of interconnected devices. This links together practical considerations when deploying quantum cryptography to electricity networks.

Grice et al [14] have investigated the applicability of quantum key distribution within cyber security systems on smart grids. The research considers different applications within smart grids and how suitable QKD is for protecting communications. Different security requirements are discussed and how quantum communication schemes must be considered with real-world requirements. This shows how quantum cryptographic techniques must be considered with the infrastructure requirements.

This shows that there are many factors to consider when considering security within smart grids. Authentication, confidentiality, communications overhead and capabilities of the participants are all factors to consider. This provides a background to look at semi-quantum methods to reduce the quantum requirements of certain participants.

2.4 Recent Research on Semi-Quantum Authentication Protocols

Zhang et al.[15] introduced a semi-quantum two-party authentication protocol for smart grids. This was focused on two parties having limited quantum capabilities. The research relates to smart grid due to the need for reliable verification of parties communicating.

The main research referenced in this project is from Lou et al.[1]. It was published in the IEEE Transactions on Dependable and Secure Computing in 2025. The authors introduced a semi-quantum private bidirectional communication protocol for smart grids. Unlike many other pieces of research, this was focused on transmitting sensitive information in a bidirectional manner. It also includes mutual authentication of identities between the parties.

The protocol proposed utilises multiple phases of communication; quantum state preparation, distribution and eavesdropping detection, mutual identity authentication, semi-quantum bidirectional communications and mutual message authentication. A key aspect of their solution is splitting quantum states into teleportation and measurement qubits. This ensures bidirectional data can be sent.

They also provided a security analysis for various types of attacks and simulated the protocol using IBM quantum cloud platform. They utilised their results to analyse how effective their

communication protocol is. This research is very important to this project due to combining quantum communication, identity verification and message authentication in a smart grid.

2.5 Comparative Analysis of Existing Research

The referenced papers look at various parts of the topic of quantum cryptography and secure communications. Some focus on the principles of Quantum Key Distribution (QKD) whilst others investigate how limited quantum devices can be used, authentication mechanisms and real world implementations in an electrical grid.
Table 2.1 gives a summary of these key research papers, what major contributions have been made by these papers and how they relate to the current project. The papers cover QKD, entanglement based communications, semi-quantum methods and authentication.

Table 2.1: Overview of Key Research Contributions in Quantum Cryptography and Smart Grid Communication

Research work	Main contribution	Relevance
Bennett and Brassard [2]	Introduced BB84 quantum key distribution	Basis for the BB84 experiment
Ekert [3]	Introduced entanglement-based QKD	Background for entangled-state operations
Boyer et al. [6]	Introduced semi-quantum key distribution	Foundation for restricted quantum operations
Iqbal and Krawec [11]	Surveyed semi-quantum cryptographic protocols	Broader understanding of semi-quantum approaches
Qasaimeh et al. [12]	Reviewed smart grid authentication techniques	Authentication requirements
Kong [13]	Reviewed QKD in smart grid communication security	Application of quantum cryptography to smart grids
Grice et al. [14]	Examined QKD applicability to smart grid cybersecurity	Practical application considerations
Zhang et al. [15]	Proposed semi-quantum two-way authentication for smart grids	Semi-quantum authentication
Lou et al. [1]	Proposed privacy-preserving bidirectional data transmission with mutual identity and message authentication	Primary foundation of this project

This demonstrates that the field of research has evolved from simple quantum key distribution to specific communication protocols capable of meeting complex requirements such as authentication and two-way communication. The work by Lou et al. [1] is particularly relevant as it considers these requirements in the context of a semi-quantum smart grid communication protocol.

2.6 Research Gap and Relevance to the Present Project

It can be seen from the literature review that there has been a significant amount of research conducted into Quantum Key Distribution, Semi-Quantum Communication and Smart Grid Authentication. Understanding these protocols requires looking at how each individual piece works and how it could be implemented using quantum circuits.

The main research paper [1] describes a full protocol and security analysis. However, in this project we have taken a simulation based approach. Looking at some of the key pieces of this research including; preparing and measuring Bell states, some authentication processes, two-way binary data sharing, and XOR based authentication. All of this has been implemented using Qiskit and Qiskit Aer.

There is also a separate Intercept and Resend attack on a BB84 protocol which helps demonstrate how eavesdropping affects quantum communication.

This aims to provide a link between the theory in the literature and how it can be implemented practically using quantum circuit simulation. However, this has been limited to the chosen pieces of the protocol and does not aim to fully implement the protocol from [1]. A formal proof of security has also not been provided.

Problem Definition and Objectives

3.1 Problem Definition

Smart grid communications involve sensitive information being shared between different parties (such as service providers and smart meters). This needs to be secured from unauthorised access, impersonation and modification. Due to advances in quantum computing, there is a demand for alternative methods to be considered for securing communications.
The purpose of this project is to investigate and simulate some of the methods used in semi-quantum communications. This will allow secure two-way communication and authentication between two parties. This project looks at combining operations on quantum states with classical processes to carry out identification, information sharing and message authentication through quantum circuit simulations.

3.2 Objectives

The major objectives of this project are:

1. **Study of Semi-Quantum Communication:** To understand the principles of semi-quantum cryptography, quantum entanglement, Bell states, and the role of participants having different levels of quantum capabilities.
2. **Implementation of Authentication Mechanisms:** To construct quantum circuits using Qiskit for Bell-state preparation and measurement, and simulate selected mutual identity authentication procedures between communicating entities.
3. **Bidirectional Data Transmission and Message Authentication:** To implement the exchange of binary information in both communication directions and explore XOR-based message authentication for verifying the consistency of exchanged messages.
4. **Simulation and Performance Analysis:** To examine the behaviour of selected protocol mechanisms through quantum circuit simulation and conduct a separate BB84-style intercept-and-resend experiment to observe measurement-related disturbances and analyze the resulting outcomes.

3.3 Project Scope

This project uses Python, Qiskit and Qiskit Aer to simulate certain protocols of quantum communications and authentication. We focused on constructing quantum circuits, measuring, authenticating identities, sending information back and forth and verifying messages. This is not an exact replication of the protocol given in the original research, nor does it offer any formal proof of security.

Proposed Methodology and Work Description

4.1 Overview of the Proposed Methodology

This project will simulate various elements of semi-quantum communication to allow secure data transmission and authentication. This has been done using the method outlined in [1] by Lou et al. to understand how quantum states can be prepared, measured, authenticated and data sent back and forth, as well as message authentication.
Python has been used to implement this project. Qiskit was used to create quantum circuits and then Qiskit Aer was used to simulate them.
The following steps have been taken:

1. Create the simulation environment and quantum circuits.
2. Create and measure Bell states.
3. Investigate decoy state verification and authentication.
4. Implement two-way communication.
5. Authentication of messages and simulation results.

An additional experiment was carried out to investigate the effects of eavesdropping using a BB84 style intercept and resend experiment.

4.2 System Design and Communication Model

This project looks at a two-party communication problem between two entities, SP (service provider) and SM (smart meter). This will examine specific quantum operations and authentication processes between the parties, using an exchange of binary data.
The simulator uses quantum operations alongside classical operations. Quantum circuits are used to create and measure quantum states, whilst classical operations are used to send binary data, authenticate and process the data.
The main parts of this system are:
Quantum State Preparation – Create quantum circuits to prepare the required quantum states, including bell states.
Measurement & Verification – Measure quantum states to obtain classical results. Specific verification processes can be examined.
Authentication – Simulate specific processes to verify identity of the entities.
Binary Data Exchange – Binary data can be exchanged between SP and SM.
Message Authentication – Use XOR processes to examine message consistency.
These components are implemented as separate but related operations within the simulation.

4.3 Preparation of the Simulation Environment

To begin, I set up the necessary programming environment. I have used Python along with Qiskit to build quantum circuits.
The technology used throughout this implementation includes:

- **Python** – Used throughout this implementation.

- **Qiskit** – Used to design quantum circuits.
- **Qiskit Aer** – Used to run quantum circuits.
- **Jupyter Notebook** – Used for development.

Once the environment has been set up, I import the necessary libraries. From here I can initialize my quantum circuits. I decide how many qubits and classical bits I would like to use depending on what operation I am simulating. Then I define my quantum gates, measurement and classical operations. I can now run my circuits using the simulator. This will give me the measurement counts which I can look at to see how my operations have behaved.

4.4 Bell-State Preparation and Measurement

A key component of the implementation is the creation and measurement of entangled quantum states. Bell states are maximally entangled states on two qubits. They have correlated measurement results.

The quantum gates are used to create the desired Bell state configuration. A Hadamard gate is applied to create superposition of the qubits. Then a CNOT gate is used to create the entanglement.

The general circuit operations are:

1. Initialize the qubits in the computational basis state.
2. Apply a Hadamard gate to the control qubit.
3. Apply a CNOT gate between the control and target qubits.
4. Measure the qubits to obtain classical outcomes.

The Bell state commonly generated using these operations is:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{4.1}$$

The resulting state produces correlated measurement outcomes when both qubits are measured in the computational basis. This behaviour is examined through repeated circuit executions.

Bell-state preparation and measurement provide the quantum foundation for studying selected authentication and communication operations in the project.

4.5 Decoy-State Verification and Identity Authentication

This part considers decoy state verification. This looks at how perturbations of the quantum states can be used to detect possible unauthorised interference. This can be done using certain quantum states for verification in quantum communications.

The project includes decoy state verification by sending and measuring specific quantum states using quantum circuits. The results of measurements are analysed to understand how these states behave in the simulated conditions.

Authentication is also an important aspect of secure communications. Entities should be able to authenticate one another before exchanging any sensitive information.

In the simulation certain protocols for authentication have been implemented using the creation and measurement of particular Bell states. The measurement outcomes are then used to determine if the conditions for authentication are satisfied.

This demonstrates how quantum state correlations can be used for authentication. However this is a simplistic example and does not fully reflect the distributed identity authentication protocol outlined in the original research paper [1].

4.6 Bidirectional Data Transmission

The following step is the exchange of binary data between both entities. Here, it is possible to send information in both directions.
There are 2 ways of communicating:

A. Service Provider to Smart Meter (SP → SM)

In this direction, the service provider will send the smart meter a binary message. This can be simulated using quantum circuits and classical computation.
Overall the steps are:

1. Define the binary message to be sent.
2. Create the correct quantum circuits.
3. Run the circuits on the quantum simulator.
4. Measure the results.
5. Process the results to get the information that was sent.

B. Smart Meter to Service Provider (SM → SP)

The smart meter sends binary information back to the service provider. This is simulated using a combination of quantum circuits and classical computation.
This measurement information is then processed to determine whether the binary information we would expect can be retrieved.
Both directions are considered individually to understand how the communication tools behave and confirm that we are able to retrieve our desired information in an ideal simulation scenario.

4.7 Message Authentication Using XOR Operations

Message authentication has been implemented to check whether the information is consistent and correct. The receiver (or senders if there are multiple) should be able to confirm that the message meets the authentication requirement.
XOR (Exclusive OR) based message authentication was implemented. Classic bitwise operations were performed. XOR will output a 1 if the two input bits are different, otherwise it will output a 0. A truth table can be seen below in Table 4.1 showing all possible combinations of input bits.

Table 4.1: Truth Table Representing the XOR Operation

Input A	Input B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

The XOR operation is applied to the relevant binary values during the message authentication procedure. The generated authentication values are examined against the expected values to determine whether the implemented verification conditions are satisfied.

The system provides separate means of authentication for each direction of communication between the SP and SM. This allows the simulation to model the message verification that takes place for SP to SM and SM to SP communications.

The XOR-based mechanism in this project represents a simplified authentication component. It should not be interpreted as a complete cryptographic message authentication code providing security against active attackers.

4.8 BB84 Intercept-and-Resend Experiment

In addition to the semi-quantum communication mechanisms, a separate BB84-style intercept-and-resend experiment is implemented to study the effect of unauthorized measurement.

In this experiment, a sender prepares quantum states using different bases. An intermediate party attempts to intercept the transmitted states, measures them using selected bases, and sends replacement states to the receiver.

The general procedure consists of the following steps:

1. The sender prepares quantum states using randomly selected bases.
2. The prepared states are passed through the communication channel.
3. An interceptor measures the states and prepares replacement states based on the measurement outcomes.
4. The receiver measures the received states.
5. The measurement results are compared to examine the errors introduced during the interception process.

The experiment demonstrates that measurements performed in an incorrect basis can disturb quantum states and introduce errors in the received information.

This experiment is conducted separately from the main bidirectional communication implementation. Its purpose is to provide an additional demonstration of measurement-related disturbance in quantum communication rather than to represent the complete eavesdropping-detection mechanism of the original smart grid protocol.

4.9 Simulation and Result Evaluation

After implementing the individual components, the quantum circuits are executed using Qiskit Aer. The simulation results are obtained from measurement counts and the corresponding classical outputs.

The evaluation focuses on the following aspects:

- **Bell-State Measurement:** Examining the measurement outcomes obtained from prepared entangled states.
- **Identity Authentication:** Checking whether the simulated authentication conditions are satisfied.
- **Data Recovery:** Comparing the transmitted binary information with the recovered information in both communication directions.

- **Message Authentication:** Examining the generated XOR-based authentication values and their verification outcomes.
- **Interception Experiment:** Observing the errors introduced during the separate BB84-style intercept-and-resend experiment.

The circuits are executed under controlled simulation conditions so that the expected behaviour of the implemented mechanisms can be examined. The results are then used to discuss the working of the selected protocol components and their limitations.

4.10 Summary of the Methodology

The proposed methodology combines quantum circuit operations and classical processing to study selected mechanisms of semi-quantum communication. Bell-state preparation and measurement form the basis for examining quantum correlations, while selected identity authentication procedures and bidirectional data exchange demonstrate the communication aspects of the implementation. XOR-based message authentication is used to examine message verification, and a separate BB84-style experiment demonstrates measurement-related disturbances.

This is an implementation that uses simulation to give a better understanding of certain parts of the core research paper [1]. This is practical example of quantum communications, and whilst this does not fully replicate the protocol from the original paper, nor provide any form of security proof, it does provide a practical example of quantum authentication using Qiskit.

Proposed Algorithms

Three key algorithms have been proposed. The first handles the main two-way communication and authentication procedure. The second simulates the quantum circuits. The third analyses the effects of an intercept resend attack using another BB84 style experiment.

5.1 Algorithm 1: Bidirectional Data Transmission and Authentication

Goal: Implementation of specific steps of semi-quantum communication including checking decoy photons, mutual authentication, two-way communication and message authentication.
Input: Shared key (K_{ab}), SP's message (ω_{SP}), SM's message (ω_{SM}), length of message (n), number of shots (S) and thresholds for checking (τ_d), (τ_{id}), and (τ_{msg})..
Output: Error rate of decoy photons (e_{SM}) and (e_{SP}), authentication rates, recovery of messages, verification of MAC and protocol status.
Algorithm:
This algorithm consists of 5 steps that will complete the required tasks. These can be seen in the table below (Table 5.1).

Table 5.1: Phases of the Bidirectional Data Transmission and Authentication Algorithm

Phase	Operation	Description
I	Initialization	Select the available quantum backend, initialize the simulator, and derive the key sequences from (K_{ab}).
II	Decoy Verification	Generate decoy states, execute measurement circuits, and calculate (e_{SM}) and (e_{SP}).
III	Identity Authentication	Prepare Bell states, divide key positions into two subsets, perform measurements, and calculate authentication rates.
IV	Bidirectional Transmission	Execute separate circuits for SP-to-SM and SM-to-SP communication and decode the measurement outcomes.
V	Message Authentication	Generate XOR-based authentication values, verify the recovered messages, and determine the final protocol status.

The two key sequences are derived from the interleaved pre-shared key:

$$K_A[i] = K_{ab}[2i]$$
$$K_B[i] = K_{ab}[2i+1] \quad (5.1)$$

The number of decoy states is determined by:

$$D = \frac{n}{2} \quad (5.2)$$

The decoy verification phase calculates the error rates as:

$$e_{SM} = \frac{N_{SM}^{error}}{N_{SM}^{total}}$$
$$e_{SP} = \frac{N_{SP}^{error}}{N_{SP}^{total}} \quad (5.3)$$

where (N^{error}) represents the number of erroneous measurement outcomes and (N^{total}) represents the total number of measurements.

If either decoy error rate exceeds the threshold (τ_d), the protocol execution is terminated. Otherwise, identity authentication is performed.

The key positions are divided into two subsets:

$$A = \{i \mid K_A[i] = K_B[i]\} \quad (5.4)$$
$$B = \{i \mid K_A[i] \neq K_B[i]\}$$

Bell-state measurements are performed for Subset (A), whereas computational-basis measurements are performed for Subset (B). The resulting authentication rates are compared with the threshold (τ_{id}).

After successful identity verification, the two communication circuits are executed independently. Their measurement outcomes are processed to obtain the recovered messages:

$$\omega_{SP}^{rec} \quad (5.5)$$
$$\omega_{SM}^{rec}$$

Finally, the message authentication values are calculated using:

$$MAC_{SP} = K_A \oplus \omega_{SP} \quad (5.6)$$
$$MAC_{SM} = K_B \oplus \omega_{SM}$$

The calculated values are compared with those obtained using the recovered messages. If both message authentication rates satisfy (τ_{msg}), the protocol execution is marked successful; otherwise, it is terminated as unsuccessful.

5.2 Algorithm 2: Quantum Circuit Construction and Simulation

Objective: To construct quantum circuits for the required quantum operations and evaluate their measurement outcomes under ideal and noisy simulation conditions.

Input: Quantum-state configurations, Bell-state labels, binary messages, measurement bases, and number of shots (S).

Output: Measurement counts, decoding results, probability distributions, and simulation performance data.

Algorithm Description:

The circuit construction and simulation procedure is organized into the following components.

A. Bell-State Preparation and Measurement

The required Bell states are generated using Pauli-X gates, Hadamard gates, and controlled-NOT operations.

The Bell state used in the implementation can be represented as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (5.7)$$

Bell-state measurement is performed using a CNOT gate followed by a Hadamard gate and computational-basis measurements.

B. Communication Circuit Construction

Separate quantum circuits are constructed for the two transmission directions:

- SP-to-SM communication uses (3n) qubits and (3n) classical bits.

- SM-to-SP communication uses (2n) qubits and (2n) classical bits.

The resulting measurement outcomes are processed using the decoding expressions implemented in the notebook.

C. Ideal Simulation

The communication circuits are executed using the local AerSimulator. Different circuit configurations are examined, including circuits without correction gates, static correction gates, and dynamic correction operations.

The recovered messages and receiver-state measurement results are compared to evaluate each configuration.

D. Noisy Simulation

The FakeQuitoV2 backend is used to construct an AerSimulator with a backend-derived noise model.

The noisy simulation evaluates three separate benchmarks to examine the effect of backend-derived noise on different quantum operations. Table 5.2 summarises the benchmarks and their respective evaluation purposes, covering single-qubit measurements, Bell-state measurements, and teleportation decoding accuracy.

Table 5.2: Noisy Simulation Benchmarks and Their Evaluation Objectives

Benchmark	Purpose
Figure 6	Examine single-qubit measurements in X and Z bases.
Figure 7	Examine Bell-state measurements using BSM and Z-basis measurement.
Figure 8	Evaluate teleportation decoding accuracy for binary message states.

Measurement counts are converted into probabilities and accuracy values. The results are stored in CSV files and represented graphically for comparison.

5.3 Algorithm 3: BB84 Intercept-and-Resend Experiment

Objective: To examine the effect of an interception attempt on the measurement outcomes of BB84-style quantum communication.

Input: Number of experimental shots (S) and random seed.

Output: Error rates for experiments with and without interception.

Algorithm Description:

The BB84-style experiment is evaluated under two configurations to examine the effect of interception on quantum communication. The behavior of communication process when there is no attack and when there is an intercept and resend attack is shown in table 5.3.

Table 5.3: Configurations of the BB84 Intercept-and-Resend Experiment

Configuration	Description
Without Attack	The sender prepares a BB84 state and the receiver measures it in the original preparation basis.
With Attack	An interceptor randomly selects a measurement basis, measures the state, resets the qubit, and prepares a replacement state based on the measurement result.

Four possible quantum states are considered:

$$|0\rangle, \quad |1\rangle, \quad |+\rangle, \quad |-\rangle$$

Each experimental transmission consists of a random choice of state from the sender. The attack scenario consists of a random choice of basis (either the X or Z basis) from the interceptor. Finally, the receiver will measure the received state in the original preparation basis.

The measurement result obtained by the receiver can be compared to the expected bit value. The Quantum Bit Error Rate (QBER) can then be determined by:

$$QBER = \frac{N^{error}}{S} \times 100 \tag{5.8}$$

Where (N^{error}) is the number of erroneous results achieved by the receiver, and S is the total number of shots in the experiment. The error rates for each configuration are compared to determine how much the interception has disturbed the system.

This experiment is independent of the main semi-quantum communication protocol, however it is included to show measurement disturbance.

5.4 Mathematical Expressions Used in the Implementation

The following expressions summarize the important calculations performed in the project.

1. Key derivation

$$K_A[i] = K_{ab}[2i] \tag{5.1}$$
$$K_B[i] = K_{ab}[2i+1]$$

2. Decoy error rate

$$e = \frac{N^{error}}{N^{total}} \tag{5.9}$$

3. Identity authentication rate

$$R_{id} = \frac{N^{matching}}{N^{authentication}} \tag{5.10}$$

4. Message authentication

$$MAC_{SP} = K_A \oplus \omega_{SP} \tag{5.6}$$
$$MAC_{SM} = K_B \oplus \omega_{SM}$$

5. Message authentication rate

$$R_{msg} = \frac{N^{verified}}{N^{message}} \tag{5.11}$$

6. Quantum channel efficiency

$$\eta = \frac{8n}{8.5n} \times 100\% \tag{5.12}$$

where (n) represents the message length.

5.5 Summary

The proposed algorithms organize the implementation into three major procedures: bidirectional communication and authentication, quantum circuit simulation, and the BB84 intercept-and-resend experiment. These procedures combine quantum operations with classical processing to examine selected mechanisms of quantum communication. Their measurement outcomes and verification results form the basis for the result analysis presented in the subsequent chapter.

Proposed Flowchart / Block Diagram / DFD

6.1 System Architecture Overview

This system has been designed to emulate certain features of semi-quantum communication to securely enable bidirectional communication. The system models two entities: Service Provider (SP) and Smart Meter (SM). A quantum circuit simulation has been created to demonstrate the preparation of Bell states, quantum measurements, authentication, bidirectional communication and message authentication.
3 main components make up the overall architecture:
Service Provider and Smart Meter entities. These entities use a pre-shared binary key to generate key sequences. The Service Provider generates quantum circuits for the required Bell states and communications. The Smart Meter takes part in the communications and verifications.
Quantum simulation environment. This was created using Python, Qiskit and Qiskit Aer. This runs the quantum circuits and returns results. These results are then processed within the classical verification environment to decode the messages sent, calculate authentication values and determine if the quantum communication has been successfully completed.
The system includes a separate BB84 style intercept resend experiment. This is used to investigate how much interference there will be when an interceptor measures and resend the quantum states. This is performed separately to the bidirectional communication.

6.2 Proposed System Block Diagram

Block diagram of system initialise the entities involved in the communication. Choose what backend to use for the simulation. Input a pre-shared key and binary messages. Split the pre-shared key into two streams and utilise them for the chosen authentication and message verification operations.
Construct a quantum circuit containing the necessary Bell states and communication circuits. Construct separate communication circuits for the Service Provider to Smart Meter and Smart Meter to Service Provider communication. Run the circuits using the chosen backend. Output the measurement results to the decoding and verification module.
Verify the measurement results of the decoy state, identity authentication and message authentication. If the required conditions for verification are met display the recovered messages and authentication result. Otherwise print that the execution was unsuccessful.
Output the recovered binary messages, measurement results, authentication results and execution result.
The layout and interaction between the main parts of the system can be seen in Figure 6.1. The communication entities (Service Provider and Smart Meter) are shown along with the pre-shared key being processed. There is also a quantum circuit construction section, simulation environment, measurement, verification, message recovery and authentication.

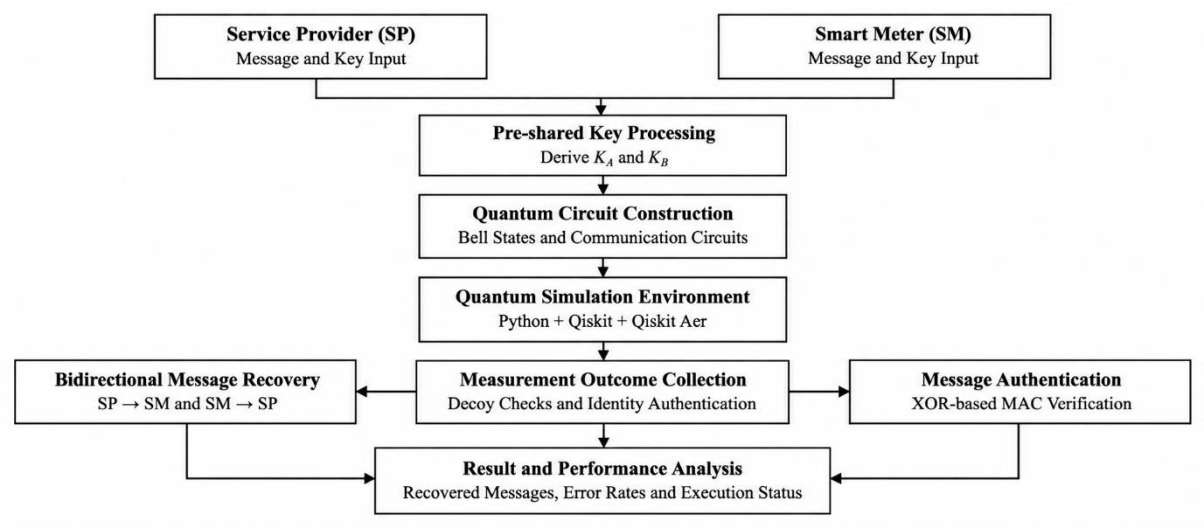


Figure 6.1: Proposed System Block Diagram

6.3 Proposed System Flowchart

The flowchart describes the sequence of operations performed by the main simulation program. It begins with system initialization and continues through the different verification and communication stages.

Step 1: Initialization

The simulation environment is initialized and the required backend is selected. The pre-shared key, binary messages, number of shots, and verification thresholds are provided as inputs. The key is divided into two sequences, (K_A) and (K_B), for use in the subsequent operations.

Step 2: Decoy-State Verification

The system will generate single circuits for verifying the decoy states. Then the error rates of SP and SM can be calculated based on the difference between expected and actual results.

If the error rate is higher than the set threshold, then it is marked as an unsuccessful run. Else the process moves on to identity authentication.

Step 3: Mutual Identity Authentication

The main positions are further split into two subgroups depending on how the matching bits from (K_A) and (K_B) interact. One subgroup will be measured using Bell state measurements and the other using computational basis measurements.

Once the results have been obtained they are compared with what is expected for the authentication process to pass. This gives us the authentication rate which can be calculated. This number is then compared with the pre-set threshold. If it does not pass the identity check then the program will stop.

Step 4: Bidirectional Data Transmission

Once the identity check is passed, the system will proceed to create and run 2 circuits for communications:

Transmitting SP to SM: The binary message from the Service Provider will be converted into a quantum circuit and executed. The results from the measurements will then be used to decode the message back into the Smart Meter.

Transmitting SM to SP: The binary message from the Smart Meter will go through a quantum circuit. The results from the measurements will be used to decode the message back into the Service Provider.

These will be tested separately in the simulation.

Step 5: Message Authentication

XOR-based message authentication codes are generated using the corresponding key sequences and original messages. These codes are then matched against those computed using the recovered messages.

Message authentication rates are determined in order to assess whether the received data meet the verification criteria.

Step 6: Final Status

If this satisfies any of the message-authentication criteria then the simulation will show a success with the reconstructed messages and authentication results. If not, it will show failure.

The whole process of the execution of the proposed system can be seen in Figure 6.2. This shows how the simulation begins with initialisation and key generation. It goes on to verify the decoy states, authenticate identities and send messages between each other. Finally message authentication is performed using XOR. The figure also outlines what decisions need to be made to decide if the simulation was successful or not.

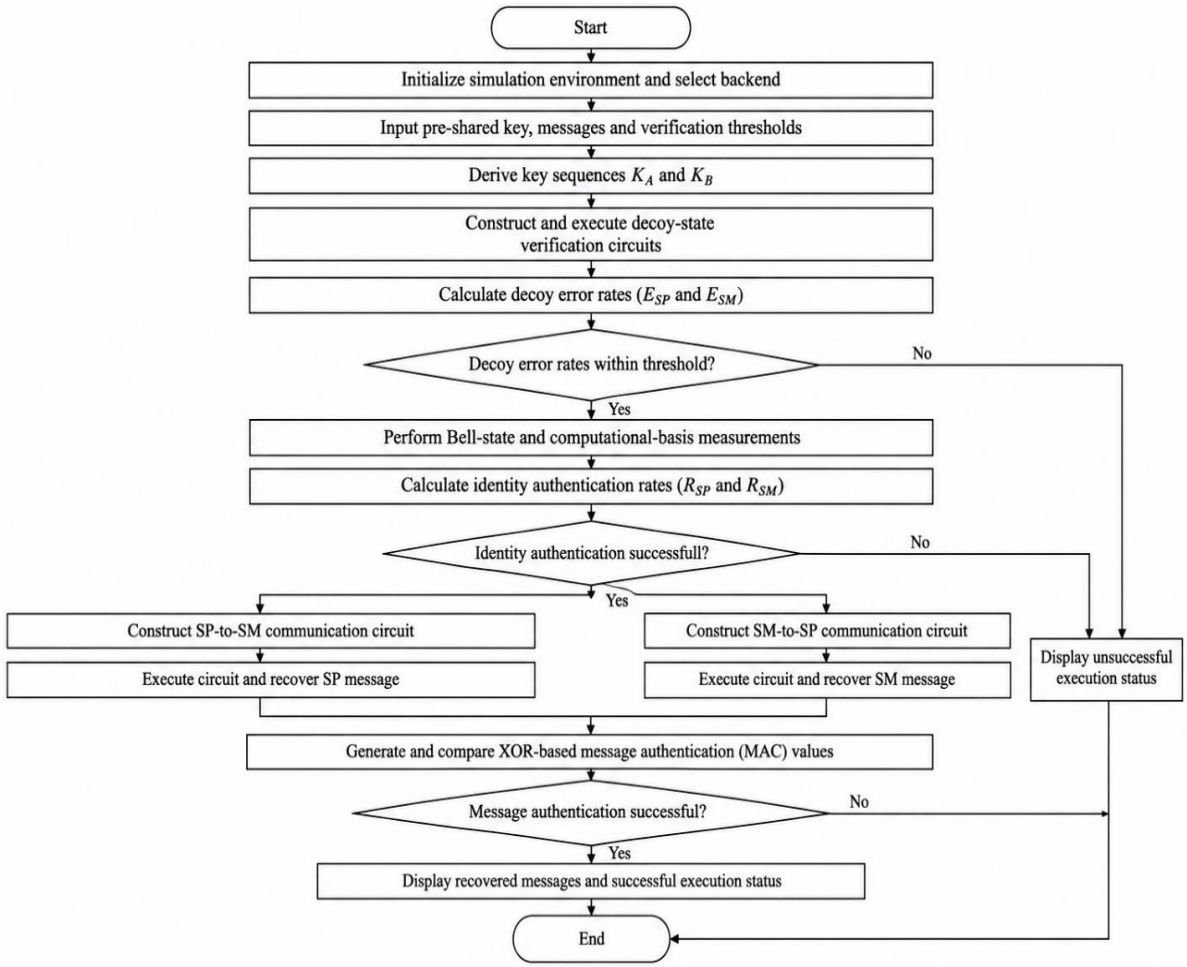


Figure 6.2: Proposed System Flowchart

6.4 Proposed Data Flow Diagram (DFD)

The Data Flow Diagram shows the flow of information between external entities and inside processes of the system. It provides details about how the input data is gathered, processed, sent through simulated quantum circuits and checked.

6.4.1 DFD Level 0 – Context Diagram

Level 0 This level shows the full simulation of the system as a single process. The Service Provider and Smart Meter are shown as two external entities providing the messages and partaking in the authentication.

The system takes in the pre-shared key, binary messages and simulation parameters. It then uses quantum circuits and classical verification methods to produce outputs such as extracted messages, measurement statistics, authentication outcomes and whether the simulation has been completed.

The interaction between the external entities and the proposed system is illustrated in Figure 6.3. The diagram represents the Service Provider (SP) and Smart Meter (SM) as external entities interacting with a single central process. It shows the exchange of input messages, pre-shared key information, communication parameters, recovered messages, and verification results.

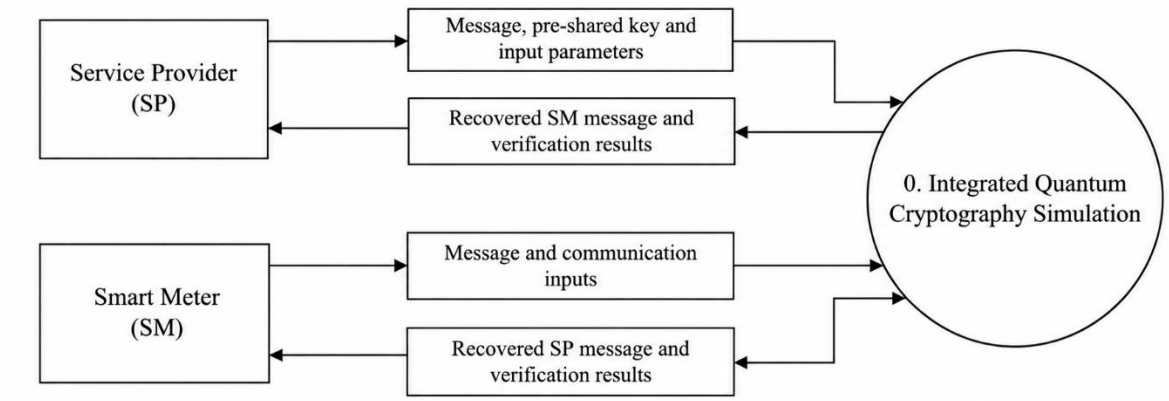


Figure 6.3: DFD Level 0 – Context Diagram

6.4.2 DFD Level 1 – Detailed Data Flow

The Level 1 DFD decomposes the main system into smaller functional processes.

1. **Input and Key Processing:** Receives the binary messages, pre-shared key, and simulation parameters. The key is divided into two sequences for subsequent processing.
2. **Quantum Circuit Construction:** Constructs the circuits required for Bell-state preparation, measurement, and bidirectional communication.
3. **Run Circuit:** Run the constructed circuits using a chosen backend and obtain measurement results.

4. **Decode & Authenticate:** Process measured results, recover any messages sent and check if authentication can be achieved.
5. **Message Authentication:** Create and compare XOR messages.
6. **Results:** Show results of any messages recovered, measured statistics, message authentication results and the outcome of the circuit run.

The DFD also models a separate BB84 style experiment. It takes quantum state configurations and experimental parameters as inputs. It then performs the intercept and resend and no attack configurations and outputs the error rates.

The detailed breakdown of the internal processes and data movement within the proposed system is illustrated in Figure 6.4. It shows how input data and configuration parameters are processed through successive stages, including quantum circuit construction, circuit execution, decoding and identity authentication, message authentication, and result generation. It also highlights the role of the data stores in maintaining key configurations and measurement outcomes.

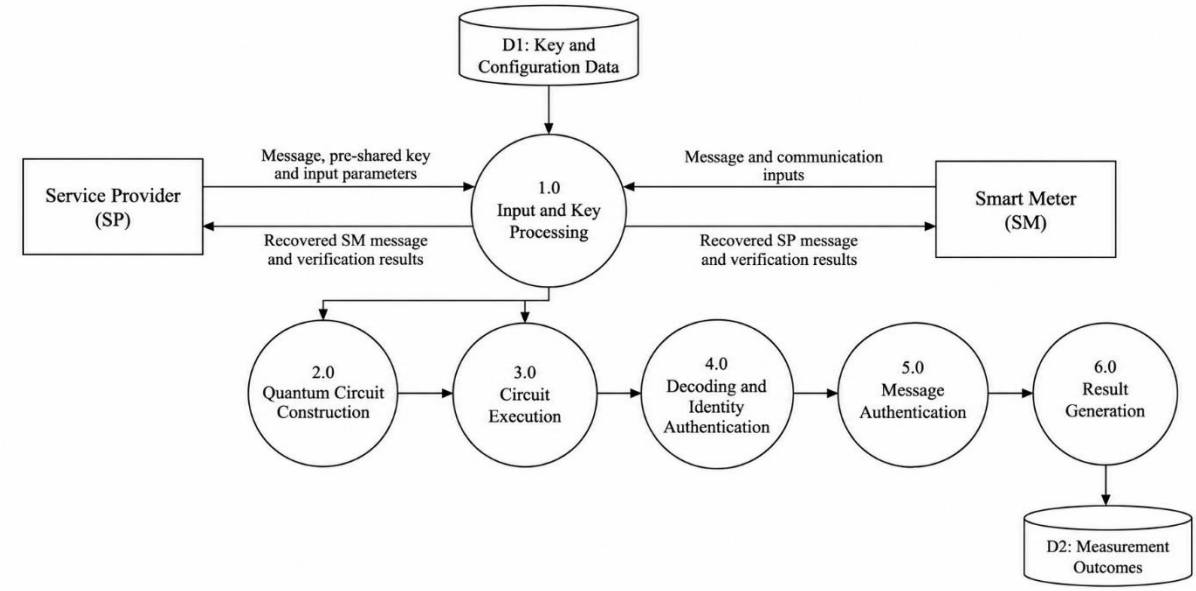


Figure 6.4: DFD Level 1 – Detailed Data Flow

Implementation (Tools and Technology Used)

7.1 Programming Language and Development Environment

This project: 'Integrated Quantum Cryptography for Secure Data and Authentication' is written using Python as it offers lots of resources for scientific computation, data analysis and quantum computing. Python has a number of libraries that can be used to create quantum circuits, run simulations, analyse data from measurements and plot graphs of the results [16].
The project was created using VS Code. The code for this project is stored in a Jupyter Notebook (bdt.ipynb). This enabled the code, text explanation, simulation results and circuit diagrams to all be in one place. This also made it easier to run specific parts of the code, view results and change parameters during the experiment [19].
A Conda environment was used to manage the packages and libraries used for this project. This provided a separate environment for the code and ensured that the correct libraries were installed. This project uses Python 3.12.14, Qiskit 2.5.2 and Qiskit Aer 0.17.2.

7.2 Qiskit Quantum Computing Framework

The main framework used for quantum computing in this project is Qiskit. This is an open source software development kit for building quantum circuits, manipulating them, compiling them and executing them. The framework allows for circuits to be created to implement circuit-based quantum algorithms. Furthermore, Qiskit offers gate based interfaces for quantum measurement and simulation [17].
In this project QuantumCircuit is used to build circuits to represent the chosen operations of the communication protocol. Quantum registers and classical registers are used to represent quantum states and measure results respectively.
The following quantum operations are key to the implementation:

- **Hadamard Gate (H):** Used to create superposition and assist in Bell-state preparation.
- **Pauli-X Gate (X):** Used to transform computational basis states, particularly to prepare the state $|1\rangle$ from $|0\rangle$.
- **Controlled-NOT Gate (CNOT/CX):** Used to create entanglement between pairs of qubits and perform operations required for Bell-state measurements.
- **Measurement Operations:** Used to obtain classical outcomes from quantum states in the computational basis.
- **Barrier Operations:** Used to visually separate different stages of a quantum circuit and make its structure easier to interpret.

For example, Bell-state preparation is implemented using a Hadamard gate followed by a CNOT gate. The resulting entangled state is:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Qiskit also provides circuit transpilation functionality. Transpilation converts a quantum circuit into a form suitable for execution on a selected backend while considering the backend's supported operations and constraints.

7.3 Qiskit Aer and Quantum Circuit Simulation

Qiskit Aer is used as the simulation framework for executing the constructed quantum circuits. Provides simulators for quantum circuits. This can model noise to see how circuits behave in an imperfect situation [18].

This uses the AerSimulator backend to simulate quantum circuits in an ideal setting. The simulator will run the circuits with a certain amount of shots. A shot is one execution of the circuit. The results will then be recorded as counts.

This will run the simulation with 8192 shots. This will give us many measurement results which we can then analyse to see if our operations are consistent.

Below shows an example of the base setup for the simulator:

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

sim = AerSimulator()
```

Once a circuit has been created, it can be transpiled for our chosen simulator, and we can run it and get some results.

```
compiled_circuit = transpile(qc, sim)

result = sim.run(
    compiled_circuit,
    shots=8192,
    seed_simulator=SEED
).result()

counts = result.get_counts()
```

The data can then be used to observe the outcomes of the measurements and see if the communications are working as expected and the authentication is happening.

7.4 Ideal and Noisy Simulation Backends

Two types of simulation conditions are considered in the project.

1. Ideal Simulation

The ideal simulation uses the AerSimulator backend, without an extra hardware noise model. This is to gain insight into the logical functionality of the circuits created. Experiments will be carried out using the preparation and measurement of Bell states, some identity verification operations, two-way communication and message verification.

The ideal simulation gives insight into whether the quantum operations performed are yielding the desired results and that the classical decoding functions can correctly decode the desired binary messages.

2. Noisy Simulation

To see the impact of imperfect quantum operations extra benchmarks have been carried out using the FakeQuitoV2 backend model. This model represents the features of the quantum device so that a noisy simulation can take place.

Noise is applied during these benchmarks to look at:

- Single qubit measurements,
- Bell state measurements,
- Teleportation decode.

This helps demonstrate the impact noise can have on measurement results and how reliable quantum operations are. The noisy benchmarks have been done separately from the main ideal communication simulation so they should not be seen as a full noisy simulation of the whole proposed communication protocol.

7.5 Supporting Python Libraries and Data Processing

There are additional python libraries used within this project to aid numerical calculations, data organisation and data visualisation.

NumPy: This library is used for numerical calculations. It allows binary sequences and simulation data to be handled. This library has many functions which allow arrays to be processed efficiently when working with measurement results and calculating values for the experiment.

Pandas: This library organises data from the simulation into tables. This helps to display how each experimental setup performed.

Matplotlib: This library creates graphs for the results of the experiment. It can create graphs comparing how well measurements performed, and noisy simulations. It can also be used alongside Qiskit circuit drawing functions to create visuals of the quantum circuits [20].

These libraries aid in converting the raw data from the simulation into tables and graphs that can be understood and added to the report.

7.6 Implementation of Authentication and Communication Experiments

Each stage of the implementation has been split into three parts: circuit design, simulation and results analysis. The first experiment to be performed was the identification verification using an 8 qubit circuit. The circuit included stages for preparing initial states, creating entangled Bell states, performing Bell state measurements on selected qubits and measuring the remaining qubit pair in the computational basis.

For the SP to SM communications experiment a 12 qubit circuit was used. The circuit prepared message based states and Bell pairs. It then performed the necessary measurements and tested different correction options.

The SM to SP communications experiment also used an 8 qubit circuit. The circuit prepared Bell pairs then performed the necessary measurements and tested the selected correction options.

Each experiment had different decoding methods, and XOR gates were used for calculating and comparing the authentication values in the message authentication part of the experiment. An additional experiment was performed for BB84 style intercept resend attack. This looked at how trying to intercept the quantum states would affect the transmission. This was done by looking at the difference in QBER values for both normal and attacked situations.

7.7 Project Repository and Code Organisation

The code and files for this project have been uploaded to a GitHub repository. In this repository you will find the Jupyter Notebook, information on dependencies, documentation for the project plus directories containing the report, presentation and results of the experiments.
The repository can be found here:
GitHub Repository: https://github.com/Naman-Aggarwal654/Integrated-Quantum-Cryptography
The repository includes the implementation and associated resources. This helps understanding how the code has been structured and what experimental was carried out.
In total, using Python, the Qiskit library, the Qiskit Aer library and other scientific libraries we have constructed an implementation that can be used to create quantum circuits, simulate selected communication protocols, perform analysis on measurement results and plot our experimental results. We have implemented selected parts of the protocols used in the reference protocol. We are not claiming that our implementation replicates the entire original protocol, nor are we providing a formal proof of security.

Result Discussion and Analysis

8.1 Experimental Setup

The proposed implementation was written in Python using the Qiskit and Qiskit Aer libraries. The main bidirectional communication and authentication protocol was implemented on the local AerSimulator, with an ideal simulation environment. 8192 shots were used in the main protocol. The simulation parameters can be seen in Table 8.1 below:

Table 8.1: Experimental Parameters

Parameter	Value
Main simulation backend	AerSimulator
Number of shots	8192
Message length (n)	4 bits
Pre-shared key (K_ab)	00110010
Derived key K_A	0101
Derived key K_B	0100
SP message (ω_SP)	0101
SM message (ω_SM)	1010
Decoy error threshold	0.20
Identity threshold	0.80
Message authentication threshold	0.80

In addition, noisy baseline circuits were implemented using a noise model extracted from the FakeQuitoV2 backend. These benchmarks were performed separately to study the effect of simulated hardware noise on quantum measurements and decoding operations.

8.2 Decoy-State Verification and Identity Authentication

The first experiment evaluated the selected decoy-state verification and identity authentication mechanisms. The decoy states are measured and compared to the threshold set.
The resulting error rates for the decoy states are:

$$e_{SM} = 0.0000, \qquad e_{SP} = 0.0000$$

Since both values are less than 0.20, execution could continue to authentication. In order to authenticate the identities of each party, we perform Bell state measurements on those positions where the key bits are the same and computational basis measurements on those positions where the key bits differ. The sets are:

$$A = \{0,1,2\}, \qquad B = \{3\}$$

The identity authenticating circuit was run for 8192 shots. This gave a 100% match success rate for both subsets.
The simulation results for verifying a decoy state and mutual identification are given in Table 8.2. Results include the calculated decoy error rates for both parties involved, match success

rates for identity within subsets A and B and an overall result for verification under ideal simulation conditions.

Table 8.2: Decoy Verification and Identity Authentication Results

Parameter	Result
SM decoy error rate	0.00%
SP decoy error rate	0.00%
Identity match rate – Subset A	100%
Identity match rate – Subset B	100%
Overall verification status	Passed

In Figure 8.1 we have a quantum circuit that demonstrates how an identity authentication protocol can be implemented by preparing and measuring in the Bell state basis. Each pair of qubits is prepared in accordance with the chosen key bit relationship. Quantum gates are applied in the form of Hadamard, controlled-Not and Pauli-X to create the desired quantum states. The measurement results are then used to determine if the conditions of identity authentication have been met.

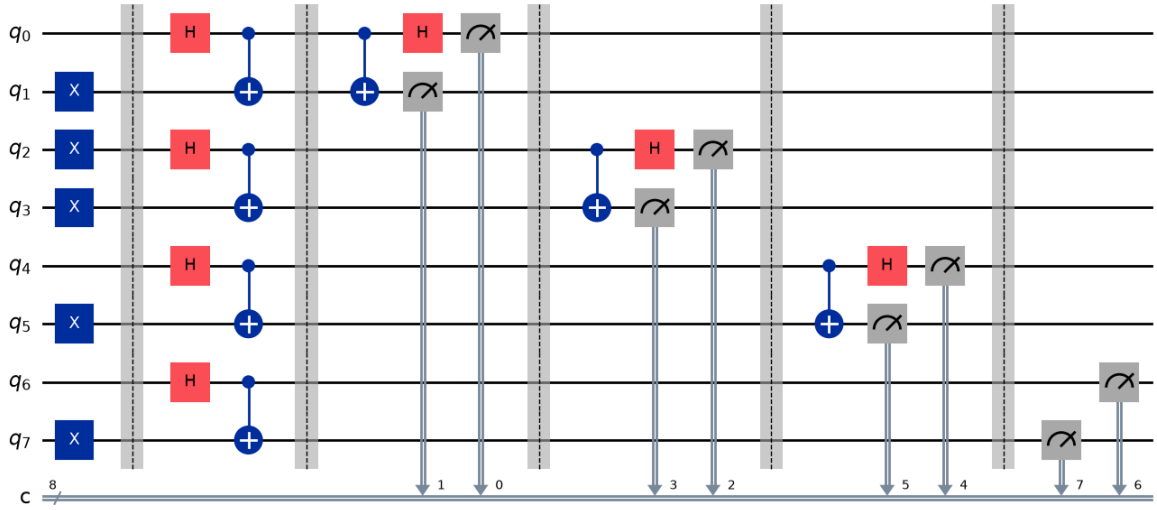


Figure 8.1: Quantum Circuit for Bell-State-Based Identity Authentication

This is a schematic diagram of the circuit implementing the combination of Bell-state operations and computational basis measurements on the two sets of authentication states. The simulation produced a 100% identity match rate for both subsets under the selected input configuration, indicating that the expected measurement conditions were satisfied.

The results indicate that the selected measurement outcomes satisfied the implemented verification conditions under ideal simulation. However, Subset B contains only one key position in this test, so its match rate should be interpreted in the context of this limited test configuration.

8.3 Bidirectional Data Transmission

The bidirectional communication experiment consisted of two separate quantum circuits: SP-to-SM transmission and SM-to-SP transmission. Both circuits were executed under ideal simulation conditions.

SP-to-SM Transmission

The Service Provider transmitted the binary message:

$$\omega_{SP} = 0101$$

The corresponding circuit consisted of 12 qubits and 12 classical bits. Three correction configurations were evaluated: without correction, static correction, and dynamic correction. Table 8.3 presents the SP-to-SM transmission results under three decoding configurations, comparing classical decoding rates and receiver-state matching accuracy.

Table 8.3: SP-to-SM Transmission Results

Configuration	Classical decoding rate	Receiver-state match
Without correction	100.00%	6.53%
Static correction	100.00%	6.74%
Dynamic correction	100.00%	100.00%

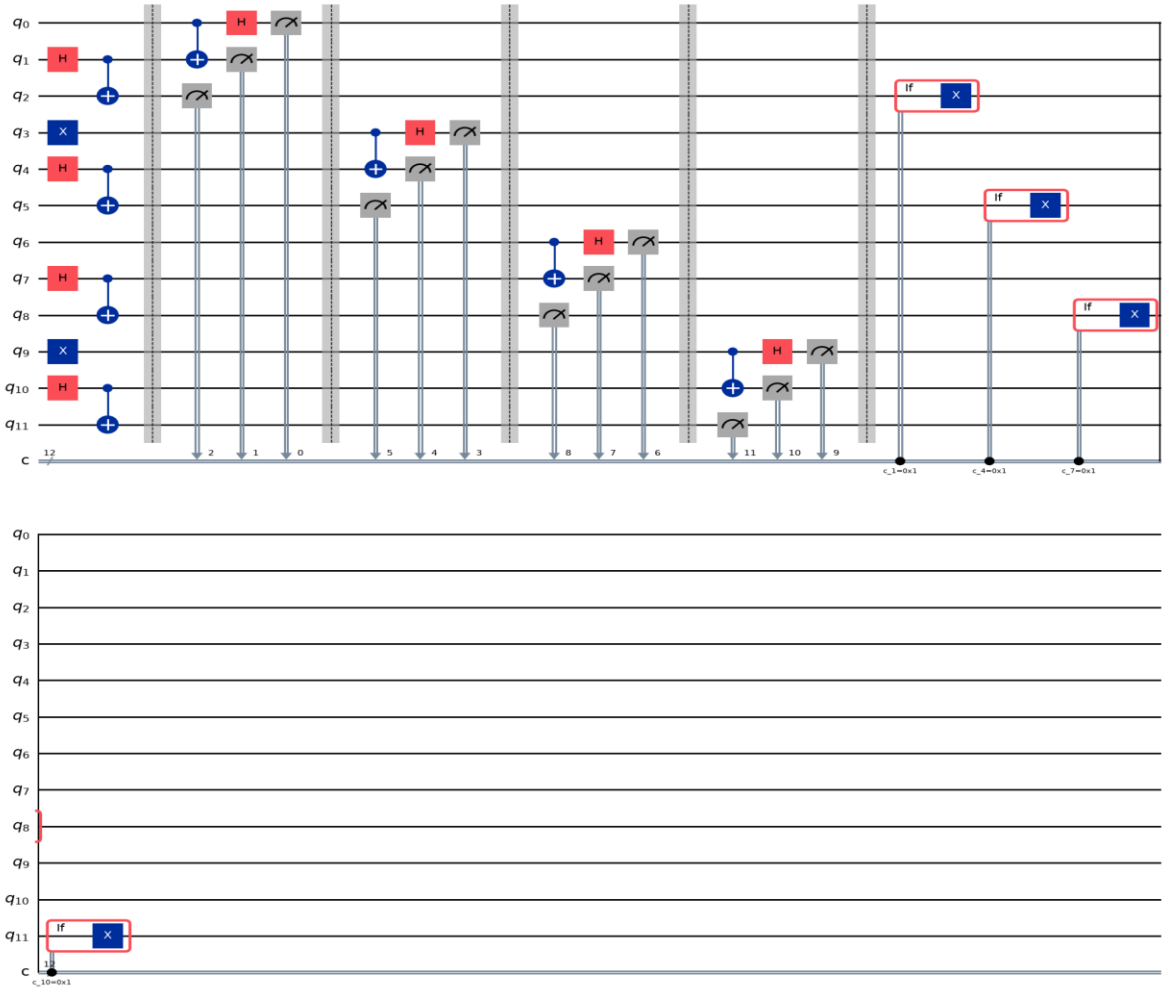


Figure 8.2: SP-to-SM Quantum Message Transmission Circuit

Figure 8.2 illustrates the quantum circuit used for transmitting the Service Provider's binary message to the Smart Meter. The circuit uses 12 qubits and 12 classical bits for a four-bit message. Quantum gates are applied to prepare the required states and perform the communication operations. Measurement outcomes are stored in classical registers and processed by the decoding procedure.

The circuit was evaluated using three correction configurations: without correction, static correction, and dynamic correction. Dynamic correction achieved 100% receiver-state matching, whereas the other two configurations produced considerably lower matching rates. The recovered message in the main protocol execution was 0101, matching the original Service Provider message.

.

SM-to-SP Transmission

The Smart Meter transmitted the binary message:

$$\omega_{SM} = 1010$$

The reverse communication circuit consisted of eight qubits and eight classical bits. The same three correction configurations were examined. Table 8.4 presents the SM-to-SP transmission results under three decoding configurations, comparing classical decoding rates and receiver-state matching accuracy.

Table 8.4: SM-to-SP Transmission Results

Configuration	Classical decoding rate	Receiver-state match
Without correction	100.00%	6.03%
Static correction	100.00%	6.08%
Dynamic correction	100.00%	100.00%

Figure 8.3 presents the quantum circuit used for communication in the reverse direction, from the Smart Meter to the Service Provider. The circuit consists of eight qubits and eight classical bits. Bell-state preparation and measurement operations are performed to support the transmission and recovery of the four-bit message.

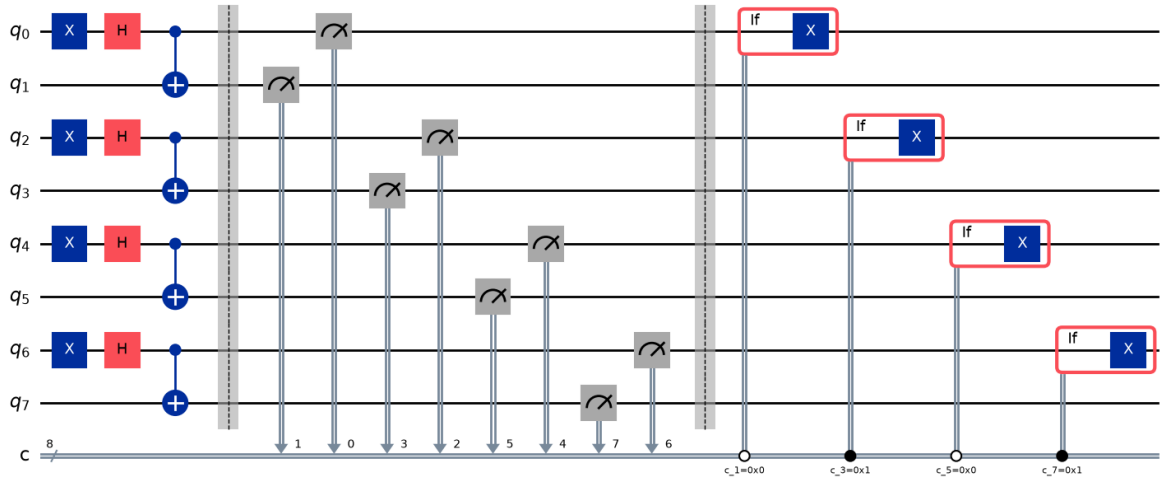


Figure 8.3: SM-to-SP Quantum Message Transmission Circuit

The circuit was tested using the same three correction configurations as the SP-to-SM experiment. Dynamic correction achieved 100% receiver-state matching, while the configurations without correction and with static correction produced lower matching rates. The main protocol successfully recovered the message 1010, which was identical to the original Smart Meter message.

.

The main protocol execution successfully recovered both messages:

Direction	Original message	Recovered message
SP → SM	0101	0101
SM → SP	1010	1010

The comparison shows that dynamic correction achieved complete receiver-state matching in both tested circuit configurations. The main protocol also recovered the expected binary messages in both directions.

8.4 Message Authentication Results

After successful identity verification and message recovery, XOR-based message authentication was performed using the derived key sequences.
The authentication values were calculated as:

$$MAC_{SP} = K_A \oplus \omega_{SP} = 0000$$
$$MAC_{SM} = K_B \oplus \omega_{SM} = 1110$$

The calculated values were compared with the corresponding values obtained from the recovered messages. Table 8.5 summarises the message authentication values, expected results, and verification status for both SP and SM communication directions.

Table 8.5: Message Authentication Results

Parameter	Result
SP authentication value	0000
Expected SP authentication value	0000
SM authentication value	1110
Expected SM authentication value	1110
SP message verification	Passed
SM message verification	Passed
Overall message authentication	Successful

Both authentication checks passed all four bit positions. Therefore, the main protocol execution completed successfully for the selected input messages and key configuration.

8.5 Noisy Quantum Circuit Benchmarks

To examine the influence of simulated hardware noise, three independent benchmarks were conducted using the FakeQuitoV2-derived noise model.

Figure 8.4 presents the measurement results obtained for four single-qubit states: $|0\rangle$, $|1\rangle$, $|+\rangle$, and $|-\rangle$. The experiment considers measurements in the computational (Z) basis and diagonal (X) basis. A noise model derived from FakeQuitoV2 was used to examine how simulated hardware imperfections influence the measurement probabilities.

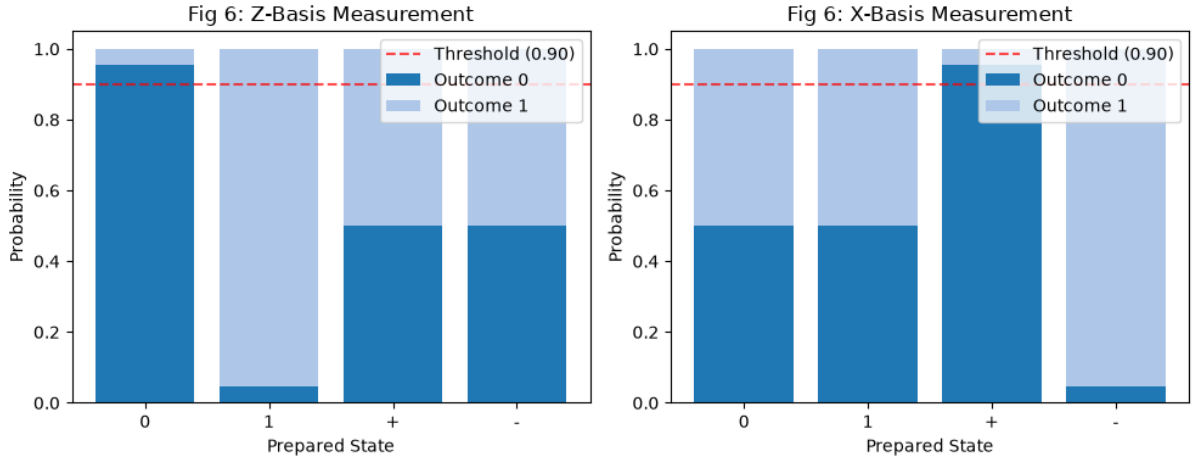


Figure 8.4: Noisy Single-Qubit Measurement Benchmark

The experiment examined the states $|0\rangle$, $|1\rangle$, $|+\rangle$, and $|-\rangle$ using X- and Z-basis measurements. As expected we saw the dominance of measurements performed in the same basis as preparation. However, there was a noticeable deviation of the measurement probabilities from their ideal values due to the simulated noise.

Figure 8.5 illustrates the measurement distributions obtained for four Bell states using Bell-state measurement and computational-basis measurement. The experiment evaluates how effectively the prepared entangled states can be distinguished under the simulated noise conditions.

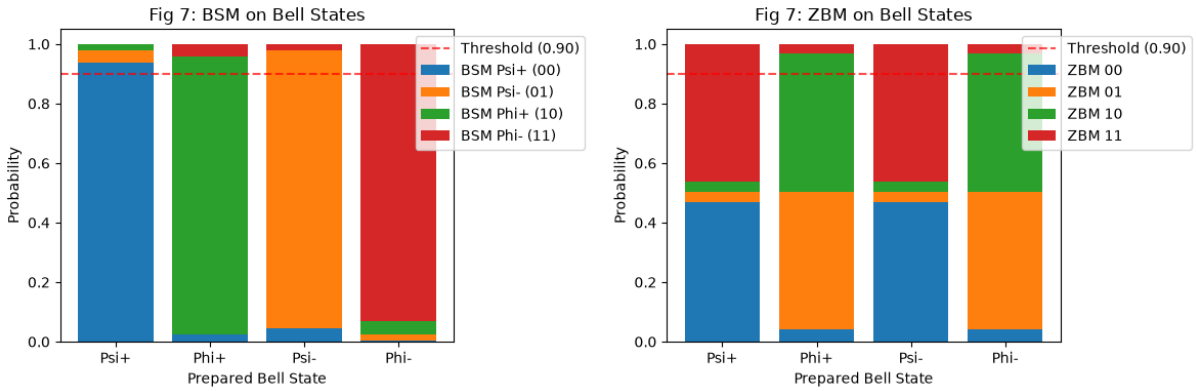


Figure 8.5: Noisy Bell-State Measurement Benchmark

Four Bell states were examined using Bell-state measurement and computational-basis measurement. The Bell-state measurement outcomes remained concentrated around their corresponding expected labels, while computational-basis measurements displayed the expected correlations rather than directly distinguishing all four Bell states.

Figure 8.6 presents the decoding accuracy obtained from the three-qubit teleportation experiment using the FakeQuitoV2-derived noise model. The experiment considers the computational-basis message states |0⟩ and |1⟩. The decoding accuracy is evaluated for each of the four possible Bell-state measurement outcomes.

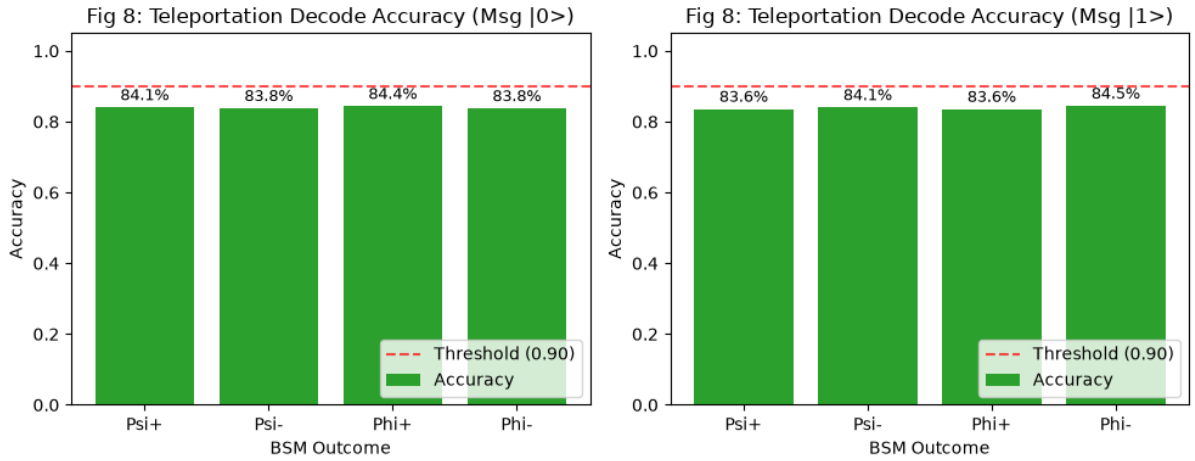


Figure 8.6: Noisy Teleportation Decoding Benchmark

The teleportation decoding experiment evaluated the message states |0⟩ and |1⟩. The observed decoding accuracies were approximately 83.6%–84.5% across the four Bell-state measurement outcomes. These results demonstrate that simulated hardware noise affected decoding accuracy compared with the ideal circuit experiments.

These three benchmarks were conducted independently and should not be interpreted as a complete noisy execution of the main bidirectional protocol.

Conclusion and Future Scope

9.1 Conclusion

This project looked at using semi-quantum cryptography to send secure information between two parties. The protocols from Lou et al. [1] were implemented using python, qiskit and qiskit aer. Quantum circuits were created that showed how to prepare and measure bell states and authenticate identities. We also successfully simulated two-way communication between the Service Provider and Smart Meter. Message encoding, decoding and authentication using XOR was also implemented. An ideal simulation was created which showed we could successfully send data and verify messages. Further simulations were completed with noise to understand imperfect quantum processes. An extra experiment was carried out that was similar to the BB84 protocol to demonstrate how measurements can cause interference. This gave an insight into quantum communications. Overall this project provides a basis for understanding quantum secure communications. However, only some mechanisms have been implemented and it is not a full re-creation of the protocol nor has it been verified to be secure.

9.2 Future Scope

Future work can be done to implement a full protocol with integrated decoy-state checking and eavesdropping detection. Realistic noise models can be investigated and implementation using real quantum hardware can be carried out. More detailed security analysis can be done to improve authentication techniques. The approach can be scaled to more complex smart grid networks with multiple service providers and smart meters.

References

[1] X. Lou, H. Zan, Z. Wang, J. Shi, and S. Zhang, "Privacy-Preserving Bidirectional Data Transmission of Smart Grid via Semi-Quantum Computation: On Mutual Identity and Message Authentication," IEEE Transactions on Dependable and Secure Computing, vol. 22, no. 5, pp. 5537–5548, 2025, doi: 10.1109/TDSC.2025.3568707.

[2] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, 1984, pp. 175–179.

[3] A. K. Ekert, "Quantum cryptography based on Bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991, doi: 10.1103/PhysRevLett.67.661.

[4] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature*, vol. 299, no. 5886, pp. 802–803, 1982, doi: 10.1038/299802a0.

[5] P. W. Shor and J. Preskill, "Simple proof of security of the BB84 quantum key distribution protocol," *Physical Review Letters*, vol. 85, no. 2, pp. 441–444, 2000, doi: 10.1103/PhysRevLett.85.441.

[6] M. Boyer, D. Kenigsberg, and T. Mor, "Quantum key distribution with classical Bob," *Physical Review Letters*, vol. 99, no. 14, Art. no. 140501, 2007, doi: 10.1103/PhysRevLett.99.140501.

[7] V. Scarani, H. Bechmann-Pasquinucci, N. J. Cerf, M. Dušek, N. Lütkenhaus, and M. Peev, "The security of practical quantum key distribution," *Reviews of Modern Physics*, vol. 81, no. 3, pp. 1301–1350, 2009, doi: 10.1103/RevModPhys.81.1301.

[8] National Institute of Standards and Technology, *Guidelines for Smart Grid Cybersecurity*, NISTIR 7628 Revision 1, Gaithersburg, MD, USA, 2014.

[9] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997, doi: 10.1137/S0097539795293172.

[10] H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," *Nature Photonics*, vol. 8, pp. 595–604, 2014, doi: 10.1038/nphoton.2014.149.

[11] H. Iqbal and W. O. Krawec, "Semi-Quantum Cryptography," arXiv preprint arXiv:1910.05368, 2019.

[12] M. Qasaimeh, R. Turab, and R. S. Al-Qassas, "Authentication techniques in smart grid: A systematic review," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 17, no. 3, 2019, doi: 10.12928/telkomnika.v17i3.11437.

[13] P. Y. Kong, "A Review of Quantum Key Distribution Protocols in the Perspective of Smart Grid Communication Security," *IEEE Systems Journal*, vol. 16, no. 1, pp. 41–54, 2022, doi: 10.1109/JSYST.2020.3024956.

[14] W. Grice, M. Olama, A. Lee, and P. G. Evans, "Quantum Key Distribution Applicability to Smart Grid Cybersecurity Systems," *IEEE Access*, vol. 13, pp. 17398–17413, 2025, doi: 10.1109/ACCESS.2025.3533942.

[15] Q. Zhang, K. Zhang, K. Hou, and L. Zhang, "A New Semi-Quantum Two-Way Authentication Protocol between Control Centers and Neighborhood Gateways in Smart Grids," *Entropy*, vol. 26, no. 8, Art. no. 644, 2024, doi: 10.3390/e26080644.

[16] Python Software Foundation, "Python 3.12 Documentation," 2023. [Online]. Available: https://docs.python.org/3.12/

[17] A. Javadi-Abhari et al., "Quantum computing with Qiskit," arXiv preprint arXiv:2405.08810, 2024. [Online]. Available: https://arxiv.org/abs/2405.08810

[18] Qiskit Development Team, "Qiskit Aer: High-performance quantum circuit simulation," Qiskit Aer Documentation. [Online]. Available: https://qiskit.github.io/qiskit-aer/

[19] T. Kluyver et al., "Jupyter Notebooks—A publishing format for reproducible computational workflows," in *Positioning and Power in Academic Publishing: Players, Agents and Agendas*, IOS Press, pp. 87–90, 2016, doi: 10.3233/978-1-61499-649-1-87.

[20] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007, doi: 10.1109/MCSE.2007.55.

Plag Report

Saurabh Jain

Plag 2

Post Quantum Cryptography Papers

Security - 2026

Indian Institute of Information Technology BP

Document Details

Submission ID

trn:oid:::1:3696833727

Submission Date

Oct 31, 2026, 9:42 PM GMT+5:30

Download Date

Oct 31, 2026, 9:45 PM GMT+5:30

File Name

tum_Cryptography_for_Secure_Data_and_Authentication_original.pdf

File Size

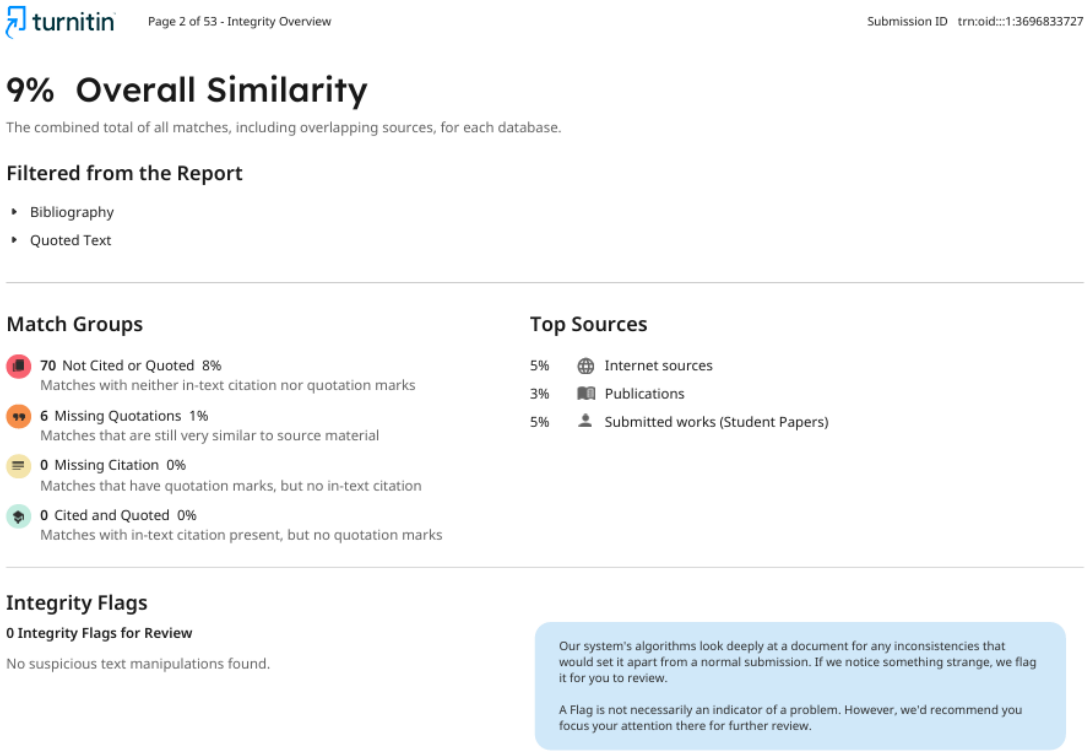
1.5 MB

47 Pages

11,792 Words

69,615 Characters

Similarity Plagiarism Report



turnitin Page 2 of 53 - Integrity Overview Submission ID trn:oid:::1:3696833727

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

70 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

6 Missing Quotations 1%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
3% Publications
5% Submitted works (Student Papers)

Integrity Flags

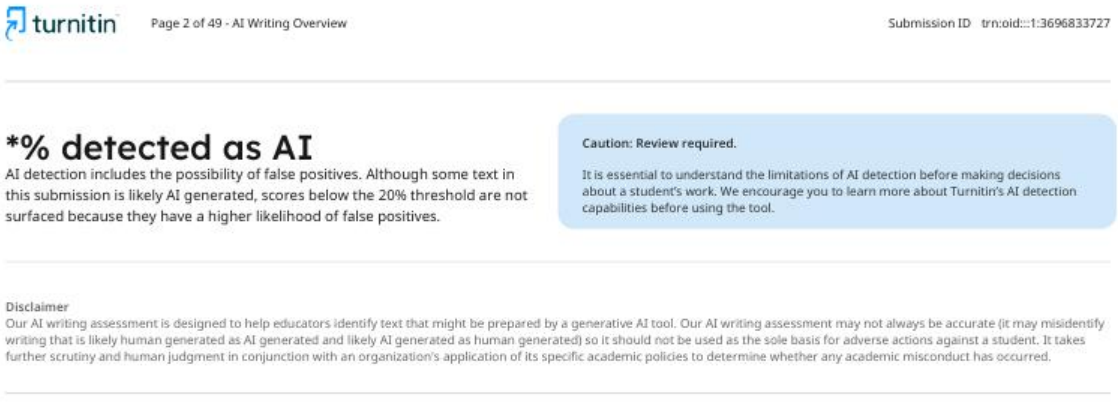
0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

AI Plagiarism Report



turnitin Page 2 of 49 - AI Writing Overview Submission ID trn:oid:::1:3696833727

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.